

CYBERSECURITY INTERNSHIP PROJECT

CyberProAI Summer Internship 2026

Ethical hacking, threat evaluation, and operational validation methodologies for AI-integrated infrastructure.

STUDENT

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INTERNSHIP SITE

CyberProAI

INSTRUCTOR

Prof. Rami Salahieh

COURSE

Cybersecurity Internship

INSTITUTION

Ivy Tech Community College

SEMESTER

Summer 2026

ABOUT CYBERPROAI

Company Overview

CyberProAI is a leading-edge cybersecurity firm providing enterprise defense architecture and advanced AI automation integrations.

Intern Role: Engaged in ethical hacking, hands-on penetration testing, systematic vulnerability scoring, and operational infrastructure documentation.

Project Purpose & Objectives

- **Classroom Integration:** Bridged academic networking and cryptography theory with corporate security tasks.
- **Tool Competence:** Learned command of industry standard reconnaissance and exploitation matrices.
- **Employability Assets:** Structured a professional repository and structured security analysis logs.

INTERNSHIP ACTIVITIES

>_ Technical Execution

- **Cympire Platform Labs:** Conducted simulated cyber warfare exercises in real-world environments.
- **Target Exploitation:** Performed network scanning, directory traversal, and host compromise scenarios.
- **BINA AI Operations:** Handled AI-augmented incident triaging and log validation tasks.
- **Infrastructure Audits:** Managed remote terminal protocols and localized system configurations.

Professional Operations

- **Stakeholder Synchs:** Participated in weekly pipeline and methodology evaluation meetings.
- **Artifact Documentation:** Created strict compliance and operations logs describing system steps.
- **GitHub Deployment:** Documented lab achievements via a public professional engineering portfolio.
- **Career Planning:** Refined resume, cover documents, and underwent live mock interview cycles.

TOOLS & TECHNOLOGIES



Exploitation Stacks

Standard systems utilized for targeted system enumeration, identification, and exploit execution.

Kali Linux

Nmap Scanner

Metasploit

Base64



Environments

Virtual hypervisors and commercial cyber ranges representing secure operational networks.

Virtual Machines

Cympire Range

GitHub

BINA AI



Methodologies

Core system vectors studied, researched, and executed throughout target discovery and report generation.

CVE Research

Reconnaissance

Privilege Escalation

Documentation

CHALLENGES & SOLUTIONS

⚠️ Core Roadblocks

- **Exploit Optimization:** Overcame initial difficulties pinpointing valid Metasploit auxiliary vectors.
- **Obfuscated Data:** Encountered hidden directories and encoded Base64 keys within the range.
- **Deliverable Backlogs:** Managed tight schedules across multiple weekly technical submissions.
- **AI Adaptability:** Acclimated rapidly to emergent security orchestration features inside BINA.

✅ Applied Solutions

- **Rigorous Research:** Conducted vendor document analysis and validated security advisories.
- **Cooperative Mentorship:** Leveraged cohort and lead reviews during operational check-ins.
- **Incremental Lab Practicing:** Re-ran lab structures in physical test rigs to reinforce techniques.
- **Process Automation:** Structured standardized report templates to accelerate weekly delivery.

CORE COMPETENCIES

Technical Skills

- **Defensive Auditing:** Adept in identifying misconfigurations and mapping exploit paths.
- **Recon & Scanning:** Competent in active scanning and live port categorization via Nmap.
- **Threat Enumeration:** Proficient in searching vulnerability repositories (CVE) for active bugs.
- **CLI Management:** Fluent in Linux commands, file permissions, and path navigation.

Professional Skills

- **Structured Technical Writing:** Able to explain highly complex findings to non-technical users.
- **Ecosystem Portfolio Control:** Experienced with Git workflows to present verification logs.
- **Collaborative Coordination:** Active in diverse task groups utilizing corporate sync procedures.
- **Critical Troubleshooting:** Systematic problem decomposition when targets fail to compromise.

RESULTS & CAREER IMPACT

KEY DELIVERABLES ACHIEVED

- ✓ **Completed Labs:** Executed active compromise routines in dynamic ranges.
- ✓ **BINA Integration:** Verified telemetry outputs with artificial intelligence.
- ✓ **Documentation Mastery:** Authored standard reports detailing operational exploits.
- ✓ **Interactive Portfolio:** Created a highly organized personal GitHub showcase.
- ✓ **Optimized Resume:** Tailored personal accomplishments for recruitment gates.
- ✓ **Linux Proficiency:** Mastered navigation and permission tasks.

Target Placement Tracks

This internship successfully contextualized system security mechanics, building a solid path for immediate market integration across security pipelines.

Cybersecurity Analyst

Junior Pen Tester

SOC Analyst

Vulnerability Analyst